

Digital Twin Technology for Sustainable Smart Cities : A Systematic Literature Review

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Abstract

Cities are witnessing a growing pressure due to expanding population and its direct and indirect pressure to cut resources and emissions that is harming the overall environment. Digital Twin Technology (DTT) has emerged as a significant enabler of smart city developments, which is a updated virtual replica of physical system, that offers a dynamic, data-driven, virtual representation of urban systems, that continuously support monitoring, simulation and evidence-based decision-making. This technology has been put forward to manage the pressure, giving policy makers and planner a testbed which addresses challenges of energy, transportation, water, waste and governance and building decision that can be simulated before they are built largely in isolation. This study presents a systematic literature review (SLR),

This paper reports a systematic literature review, carried out following PRISMA 2020 reporting guidance, of research on digital twins in sustainable smart cities. Database searches of Scopus, Web of Science, IEEE Xplore and ScienceDirect returned 214 records; after duplicate removal, title/abstract screening, and full-text assessment, 19 studies met the inclusion criteria and were analysed thematically. The review traces the concept from its origins in manufacturing and aerospace engineering through to its adaptation for urban-scale governance, and it summarises how digital twins are currently applied across energy management, transport and mobility, water and waste systems, environmental monitoring, buildings, and urban governance. It also draws together the barriers reported most consistently in the literature, chiefly interoperability, data quality and privacy, cost, and institutional capacity, and closes with directions for future research, including artificial intelligence integration and stronger empirical evaluation of sustainability outcomes.

Keywords: digital twin, technology, smart city, urban sustainability, urban digital

twin, systematic literature review, PRISMA, urban governance

Introduction

More than half of the world's population now lives in urban areas, and this share continues to climb. That trend puts steady pressure on energy grids, transport networks, water systems, and public infrastructure, and it forces city administrators to reconcile two goals that do not always pull in the same direction: improving services for residents while limiting the environmental cost of doing so. Smart city programmes have tried to address this for more than a decade by adding sensors, connectivity, and analytics to urban systems, yet a recurring criticism in the literature is that many such programmes stall at the level of isolated pilots rather than producing integrated, city-wide decision-making (Silva et al., 2018). Digital twin technology has been proposed as one way past that ceiling, offering a live, data-linked model of the city that can be tested and adjusted before changes are made to the physical environment (Bibri et al., 2024).

The digital twin concept did not originate in urban planning. It was first articulated by Michael Grieves in a product-lifecycle-management course at the University of Michigan in 2003, and it matured within NASA's aerospace programmes, where ground-based replicas of spacecraft systems were used to diagnose faults occurring in orbit (Grieves & Vickers, 2017). Over the past decade it has been rescaled from individual products and buildings to entire urban systems, producing what the literature now calls the urban digital twin or city digital twin (Deng et al., 2021; Ferré-Bigorra et al., 2022). What separates a genuine urban digital twin from an ordinary three-dimensional city model is the presence of a working feedback loop: sensor data continuously updates the virtual model, and in more advanced deployments, the outputs of that model feed back into planning and operational decisions (Mohammadi & Taylor, 2017).

The appeal of the technology for sustainability governance is that it offers a single environment in which sectors that are usually managed separately, energy, transport, water, waste, and land use, can be modelled together (Weil et al., 2023). Reviews of implemented projects, however, describe a wide spread of ambition. Some cities have built largely passive three-dimensional visualisations, while others have moved toward predictive or even semi-automated systems (Mazzetto, 2024). At the

same time, a growing body of work is documenting the obstacles that keep most digital twins from reaching that second stage, including weak interoperability between data systems, unresolved questions of data ownership and privacy, and the cost of the underlying infrastructure (Lei et al., 2023; Weil et al., 2023).

Because this literature has grown quickly and tends to be organised around single sectors or single cities, a synthesis that pulls the findings together across domains is useful both for researchers entering the field and for practitioners deciding where to focus limited resources. This review was guided by four questions:

RQ1. How has digital twin technology been defined and structured for use in sustainable smart cities?

RQ2. Which urban sustainability domains have received the most attention, and what outcomes are reported?

RQ3. What barriers are most frequently reported as slowing adoption?

RQ4. Where does the literature point for future research?

Section 2 sets out the review methodology. Section 3 covers the conceptual and technical background. Section 4 works through the sustainability domains in which digital twins have been applied. Section 5 looks at implemented case studies. Section 6 discusses the barriers reported across the reviewed studies, Section 7 identifies emerging directions, and Sections 8 and 9 close with a discussion and conclusion.

Methodology

This review followed the PRISMA 2020 reporting guidelines (Page et al., 2021), which build on the earlier PRISMA statement (Moher et al., 2009) and are now the standard reference point for systematic reviews in the digital twin and smart city literature (Afzal et al., 2023).

1. Search Strategy

Four databases were searched: Scopus, Web of Science, IEEE Xplore, and ScienceDirect. The search string combined terms for the technology ("digital twin", "urban digital twin", "city digital twin") with terms for the setting ("smart city", "smart cities") and terms for the outcome of interest ("sustainab*", "sustainable development"), joined with Boolean operators and applied to titles, abstracts, and keywords. The search was limited to English-language material published between 2015 and mid-2026, which covers the period in which the urban digital twin literature

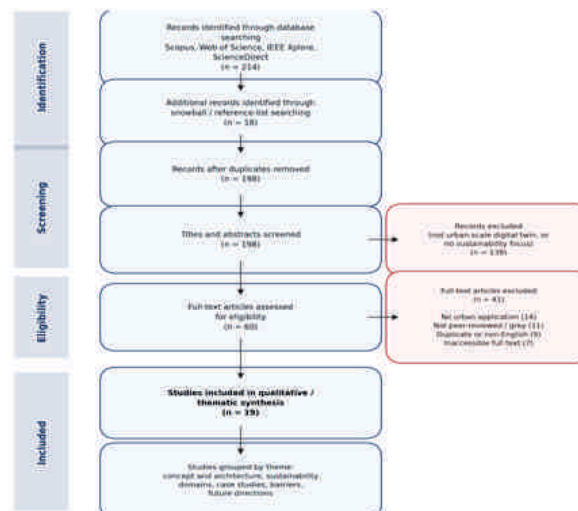
has actually taken shape; almost nothing meeting the inclusion criteria predates 2017.

2. Inclusion and Exclusion Criteria

A record was kept if it was (a) a peer-reviewed journal article, peer-reviewed conference paper, or an equivalent scholarly book chapter; (b) concerned with digital twin technology at the scale of urban infrastructure rather than a single isolated product; and (c) addressed at least one dimension of environmental, economic, or social sustainability. Records were dropped if they dealt with digital twins purely in manufacturing, aerospace, or clinical settings with no urban dimension, if they were editorials or non-peer-reviewed commentary, or if the full text could not be obtained. Reference lists of the studies that passed full-text review were also checked for further eligible sources, a step commonly used in comparable reviews to reduce the chance of missing relevant work (Weil et al., 2023).

3. Selection Process and Results

Database searching returned 214 records, with a further 18 identified through reference-list checking. After duplicate removal, 198 unique records were screened at title and abstract level, of which 138 were excluded for falling outside the scope of urban-scale digital twins or lacking any sustainability dimension. The remaining 60 were retrieved in full text; 41 were subsequently excluded, most often because they addressed non-urban applications, were not peer-reviewed, were duplicates or non-English publications, or could not be accessed in full. Nineteen studies were carried forward into the synthesis. Figure 1 summarises this process.



4. Data Extraction and Synthesis

For each included study, the review recorded the publication year, sectoral focus, methodological approach, technologies discussed, and reported outcomes or barriers. Study designs varied widely, from large bibliometric analyses (Mazzetto, 2024) to single-domain reviews (Lei et al., 2023; Weil et al., 2023), which made a quantitative meta-analysis impractical. A narrative thematic synthesis was used instead, with themes developed inductively from the extracted data and organised into the sections that follow.

Conceptual Foundations

1. Origins and Definitions

The term "digital twin" is usually credited to NASA's John Vickers, building on Michael Grieves' earlier work on mirrored virtual and physical product spaces (Grieves & Vickers, 2017). Grieves and Vickers describe a digital twin as a virtual representation of a physical asset that is detailed enough to capture its behaviour from small-scale components up to the whole system, linked to its physical counterpart by a continuous flow of data in both directions. That definition was written with manufactured products in mind, but it translates reasonably well to cities once "asset" is read as an entire urban system rather than a single machine (Deng et al., 2021).

The shift from product-level to city-level twins was made possible by two things arriving at roughly the same time: geographic information systems becoming cheap and widely usable, and building information modelling maturing within construction and architecture (Mohammadi & Taylor, 2017). Once individual buildings and infrastructure assets could be modelled digitally, the next step, stitching them together into a single city-scale representation, followed fairly naturally. Ferré-Bigorra et al. (2022) note, however, that this rapid, largely uncoordinated growth has left the field without a single agreed definition, and that implemented urban digital twins share fewer common features than the shared terminology might suggest.

2. Architecture and Enabling Technologies

Despite that definitional looseness, most descriptions of urban digital twin architecture converge on a similar three-part structure: a physical layer of sensors and infrastructure, a data layer handling transmission and storage, and an analytical layer where models and simulations run (Bibri et al., 2024). The Internet of Things

supplies the continuous sensor readings, traffic counts, energy meters, air-quality data, that keep the virtual model aligned with conditions on the ground. Building information modelling contributes detail at the individual structure level, while geographic information systems provide the spatial framework needed to integrate that detail across a wider area (Mohammadi & Taylor, 2017).

A more recent thread in the literature is the pairing of digital twins with artificial intelligence, which turns what was originally a visualisation and monitoring tool into something closer to a forecasting or decision-support system (Bibri et al., 2024; Bibri, Krogstie, et al., 2023). Cloud and edge computing supply the processing capacity such systems need, and Mazzetto's (2024) bibliometric analysis of the field notes a marked rise, from roughly 2021 onward, in publications combining digital twins with machine learning methods specifically for urban applications, rather than treating the two as separate research strands.

Application Across Urban Sustainability Domains

The studies reviewed here describe a fairly consistent set of domains in which digital twins have been applied to sustainability problems: energy, transport and mobility, water and waste, environmental monitoring, buildings, and governance (Weil et al., 2023). Each is discussed below.

1. Energy Management

Energy applications are among the more established uses of urban digital twins. By combining smart-meter data with weather and demand forecasts, a digital twin can help a utility anticipate peak load and match generation more closely to actual consumption, reducing waste in the process (Bibri et al., 2024). Bibri, Krogstie, et al. (2023) frame this within a broader argument that artificial intelligence of things applications, of which energy-focused digital twins are one instance, are becoming central to how eco-cities pursue environmental sustainability targets, rather than being treated as a peripheral technical add-on.

2. Transport and Mobility

Transport is the domain most often used to illustrate digital twin capability in the reviewed literature, largely because traffic simulation was one of the earlier and more tractable applications of the technology (Mazzetto, 2024). Reported uses include modelling traffic flow to identify congestion points, adjusting signal timing, and

supporting public transit planning. Weil et al. (2023) list mobility and road infrastructure among the domains most consistently represented in the urban digital twin literature, alongside energy and water systems, which suggests transport is not just a convenient demonstration case but a genuinely active area of ongoing research.

3. Water and Waste Management

Water systems, sewage, and drainage appear repeatedly in reviews of urban digital twin applications, generally in relation to leak detection, network monitoring, and flood forecasting (Weil et al., 2023). Waste management is discussed less often and appears to be at an earlier stage of development; where it does appear, the emphasis is on combining sensor data with routing optimisation to reduce unnecessary collection trips, which fits the broader pattern of digital twins being used first for monitoring and only later for active operational control (Ferré-Bigorra et al., 2022).

4. Environmental Monitoring and Climate Resilience

Environmental applications span air- and water-quality monitoring, pollution tracking, and flood or disaster preparedness. Ferré-Bigorra et al. (2022) describe the Herrenberg digital twin in Germany as an example built specifically around traffic-related air pollution, illustrating how a single environmental concern can drive the design of an entire local implementation rather than being added as an afterthought to a more general-purpose model. At a larger scale, Bibri et al. (2024) report a marked increase, particularly since 2020, in research combining artificial intelligence with digital twins for environmental planning, which they treat as one of the more active convergence points in the current literature.

5. Buildings and the Built Environment

At the scale of individual structures, digital twins extend building information models with live sensor feedback to support maintenance scheduling and energy performance tracking across a building's operating life. Afzal et al. (2023), reviewing digital twin applications in construction under PRISMA guidance, find that most published work still concentrates on the design and construction phases rather than long-term operation, and they argue that the operational phase, where a building's environmental footprint actually accumulates, remains comparatively under-researched. More recent work has pushed specifically into zero-energy building design, arguing that pairing artificial intelligence with digital twins at the building level can meaningfully support climate mitigation goals when scaled across a city's building

stock (Bibri et al., 2025; Bibri & Huang, 2025).

6. Urban Governance and Citizen Engagement

A smaller strand of the literature looks past the technical layer to ask what digital twins mean for how cities are governed. Deng et al. (2021) propose the idea of the "digital twin city" as a genuinely new pattern of urban governance, not simply a planning tool bolted onto existing institutional arrangements, and argue that realising this potential depends on cities also adapting how decisions get made, not just on the underlying technology. White et al. (2021) describe a digital twin platform built specifically to let residents comment on and interact with proposed developments, which points toward participation as a use case in its own right rather than a secondary benefit of transparency. On the operational side, Ford and Wolf (2020) and Wolf et al. (2022) examine digital twins built for disaster and incident management, where the twin supports coordination across multiple agencies during an active emergency rather than routine day-to-day planning.

Case Studies

A handful of implementations recur across the reviewed literature as reference points. Mazzetto's (2024) bibliometric review discusses Singapore and Dubai as examples of cities that have pursued comparatively comprehensive digital twin programmes spanning multiple urban sectors rather than a single narrow application. At the other end of the spectrum, Ferré-Bigorra et al. (2022) describe the Herrenberg twin in Germany as a much more tightly scoped project, built to model traffic pollution and mobility rather than to serve as a general-purpose city model. Between these poles sit projects such as the citizen-feedback platform described by White et al. (2021), which was designed around a specific governance function, public consultation on development proposals, rather than around a particular infrastructure sector.

Read together, these cases suggest that the field has not converged on a single template for what an urban digital twin should look like. Ferré-Bigorra et al. (2022) make this point explicitly, arguing that implemented projects share the same label but often very little else in terms of scope, data sources, or the degree to which the model can actually act back on the physical city rather than simply representing it. Most documented projects remain closer to the narrow, single-purpose end of that

spectrum than to the comprehensive, cross-sectoral end (Mazzetto, 2024).

Barriers to Adoption

Barriers to urban digital twin adoption are one of the more thoroughly documented themes in the literature. Lei et al. (2023) conducted a systematic review paired with a Delphi survey of experts, given how uneven the empirical evidence still is in places, and identified eight categories of challenge: interoperability and semantics, infrastructure (storage, computation, network capacity), data acquisition and actuation, data quality and harmonisation, modelling and simulation limitations, alongside further governance-related concerns. The main themes are discussed below.

1. Interoperability and Standardisation

Because an urban digital twin has to pull together data from legacy municipal systems, proprietary vendor platforms, and independent sensor networks, achieving technical and semantic interoperability is repeatedly flagged as one of the harder problems in the field (Lei et al., 2023; Weil et al., 2023). Ferré-Bigorra et al. (2022) argue that the absence of shared standards is a large part of why current implementations look so different from one project to the next, and they call for stronger standardisation efforts as a precondition for digital twins developed by different vendors or municipalities becoming genuinely interoperable.

2. Data Quality, Privacy, and Security

A digital twin is only as good as the data feeding it, and sensor networks are prone to gaps, drift, and noise that degrade simulation accuracy over time (Lei et al., 2023). Privacy is a separate but related concern, particularly once a twin incorporates data that could be traced back to individual households or travel patterns; Weil et al. (2023) list this among the governance-related barriers their review identified, alongside the more purely technical issues of storage and computation capacity.

3. Cost and Institutional Capacity

Building and maintaining a city-scale digital twin is expensive, in sensor deployment, cloud infrastructure, and specialist staff, and this cost falls more heavily on smaller municipalities than on major cities with larger budgets (Weil et al., 2023). This likely explains why the more comprehensive implementations documented in the

literature tend to cluster in a small number of well-resourced cities (Mazzetto, 2024), while most other projects stay narrowly scoped to a single sector or problem.

4. Governance and the Digital Divide

Beyond the purely technical barriers, several reviews raise governance questions that are harder to solve with better engineering: who owns the data a digital twin generates, which agency is accountable for decisions the twin informs, and whether the expertise needed to interpret its outputs ends up concentrated among a small group of specialists rather than shared more broadly across a city's administration (Deng et al., 2021; Lei et al., 2023). These are treated in the literature less as implementation details and more as open questions that will shape how equitably the benefits of the technology are distributed.

Future Research Directions

A few directions stand out across the more recent studies in this review. One is the continued integration of artificial intelligence into digital twin platforms, moving them from passive monitoring toward genuine forecasting and, in some cases, automated response (Bibri et al., 2024; Bibri, Krogstie, et al., 2023). A second is a sharper focus on the operational phase of buildings, an area Afzal et al. (2023) identify as comparatively neglected relative to the design and construction phases where most current digital twin research in the built environment is concentrated. A third is standardisation: several reviews call for shared data protocols and ontologies as a way of making digital twins built by different vendors and municipalities easier to compare and connect (Ferré-Bigorra et al., 2022; Lei et al., 2023).

A more general gap running through the literature is the shortage of studies that measure sustainability outcomes directly, emissions avoided, resources saved, service improvements, rather than describing technical feasibility or architectural design (Mazzetto, 2024). Much of the current evidence base documents what digital twins can in principle do; comparatively little of it documents what they have actually achieved once deployed. Closing that gap, and tying digital twin applications more explicitly to established sustainability frameworks, is likely to matter more for the field's credibility over the next few years than further architectural refinement.

Discussion

Taken together, the studies reviewed here describe a technology that has moved a reasonable distance from its manufacturing and aerospace origins but has not yet settled into a stable, agreed form for urban use. Coverage is uneven across domains, energy, transport, and water are comparatively well represented, while waste management and governance-oriented applications are thinner, and it is uneven across cities too, with a small number of well-resourced examples doing most of the work of demonstrating what a comprehensive digital twin could look like (Mazzetto, 2024; Weil et al., 2023).

That unevenness matters for the sustainability case made on behalf of digital twins. A twin that only covers one sector cannot do much to support the cross-sectoral trade-offs, energy against transport, water against land use, that real sustainability decisions usually involve (Deng et al., 2021). The barriers catalogued in Section 6 are not simply implementation friction, then; they are a large part of why the technology's more ambitious, cross-sectoral promise has been realised in only a handful of places so far.

This review has the usual limitations of a systematic review conducted in a fast-moving field. The literature search, while extending into 2026, cannot capture every recent preprint or conference paper, and the studies included vary enough in design, from large bibliometric analyses to single-city case reviews, that a narrative synthesis was more appropriate here than a quantitative pooling of results. Future reviews would benefit from more consistent outcome reporting in primary studies, which would make a more rigorous quantitative synthesis possible.

Conclusion

This review has traced digital twin technology from its origins in manufacturing through to its current, still-developing role in sustainable smart cities, covering the underlying architecture, the sustainability domains where it has been applied, a set of representative case studies, and the barriers that continue to limit wider adoption. The evidence suggests real and growing potential across energy, transport, water, and building applications, but that potential has so far been demonstrated most fully in a small number of well-resourced cities rather than across the much larger set of municipalities that might benefit from it. Closing that gap will depend less on further

technical refinement of the twins themselves and more on standardisation, clearer governance arrangements, and a stronger body of evidence showing what these systems actually deliver once they move from pilot to routine use.

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